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Monitoring and evaluation
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theory to practice

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Monitoring and evaluation of mission-oriented innovation policies: From theory to practice

Missions are nested entities involving multiple interventions at different levels and unclear and evolving boundaries, making traditional evaluation approaches ill-suited to capturing their additionality. This paper proposes mission-evaluation processes and tools that are consistent with their specific features. It notably proposes mission criteria related to their different expected systemic effects; a mission theory of action to support a developmental evaluation that tracks the evolution of the mission design and processes; and a monitoring tool to assess and compare mission-readiness levels across missions and at different stages of the mission life cycle.

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The project's overarching aim is to enhance the evidence base on national mission initiatives to provide practical inputs for effective mission implementation. The project focuses on three dimensions of mission implementation: the governance of missions, the management of mission portfolios, and the monitoring and evaluation (M&E) of missions.

A specific STI Policy Paper is dedicated to each of these three mission dimensions. This particular paper deals with the topic of the M&E of missions.

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Executive summary

Effective monitoring and evaluation (M&E) of mission-oriented innovation policies (MOIPs) helps governments and partners improve evidence-informed policy making and mission design, co-ordination and implementation, thereby enhancing economic and societal impacts. It promotes continuous learning across different stages of the mission life cycle and accountability for the significant resources invested in these policies. Even more importantly, in the current early stage where missions have yet to produce measurable impacts, M&E should help substantiate the added value of MOIPs as an additional framework in already complex and crowded policy landscapes. This is especially important because MOIPs – along with other proactive policies aimed at transforming sociotechnical systems to address societal challenges – are subject to growing criticism from some policy makers and stakeholders. The evolving complex, multidimensional and multilevel nature of MOIPs, however, makes it difficult to establish causal relationships between actions and outcomes.

Developing an evaluation framework that is valid for different types of MOIPs is challenging, due to their highly diverse types, objectives and governance structures. Evaluators must adapt their approaches to each MOIP, focusing not only on outcomes, but also on the processes and governance structures that shape these initiatives. Developing a theory of action that clarifies the hypotheses and rationales underpinning the mission design could be instrumental in this regard.

Accounting for the complex geography of MOIPs is crucial because of the ambiguous boundaries of missions, especially those with a broad systemic scope. These missions involve a wide range of actors, interventions and activities, often co-ordinated through rather loose arrangements, making it difficult to delineate their scope clearly and assess their influence. Evaluators must map out the mission's scope, identify who was engaged in its design and implementation, and assess the sphere of influence to determine how effectively the mission integrates and co-ordinates the various activities needed to fulfil the mission.

Combining the evaluation of the different nested MOIP entities: MOIPs are multilevel, nested structures where individual projects are part of portfolios within broader missions, all contributing to a larger MOIP initiative. Each entity within this hierarchy – projects, portfolios, missions – has different objectives and expected effects. This requires combining different evaluations in a tailored, co-ordinated and balanced approach. Notably, evaluations should build from the bottom up, with results at each level informing higher-level evaluations (“capillarity”). Evaluations at each level should focus on effects that cannot be captured at a lower level (“subsidiarity”).

Assessing the double additionality of missions: Due to their complex, multilevel nature, the additionality of MOIPs cannot be measured against a “no-intervention scenario” (“first-order additionality”), as is traditionally the case in public policy evaluation. Evaluators should examine a mission's impact relative to a scenario where traditional policies would have been used instead (“second-order additionality”), i.e. a counterfactual scenario with less directionality, holistic co-ordination and policy mix integration. Evaluators must also consider any potential negative effects, such as an increased administrative complexity or the displacement of new initiatives.

Embracing the evolutive nature of MOIPs: Missions often begin with broad objectives that are refined over time through negotiations, growing trust among partners and the inclusion of new stakeholders. This

ongoing evolution complicates evaluation, as both the objectives and the mission's structure are moving targets. Evaluators must assess not only the mission's outcomes but also its evolution in terms of design, governance and processes. This involves a developmental evaluation approach to track how the mission becomes more aligned with its purpose, focusing on the adaptation and learning of both policy makers and partners. Over time, as missions mature, summative evaluations can complement this process by assessing progress toward final objectives.

Adopting a nuanced approach: When evaluating MOIPs, it is crucial to recognise that more extensive orientation, co-ordination and integration are not always better. Mission partners should identify the aspects of co-ordination that add value in each specific case, taking into account the mission's context and needs. Evaluators should therefore consider the unique structural features of the challenge area addressed by each mission, such as its complexity, uncertainty and contestability, to determine the most appropriate design and avoid a one-size-fits-all approach.

The paper proposes mission-evaluation criteria, processes and tools that are consistent with the specific features of missions, as laid out above.

First, it presents **four main groups of mission effects: directionality effects, governance effects, portfolio effects and material effects**. For each of these, specific criteria are derived to guide the evaluative work (for instance mobilisation, intentionality and flexibility for directionality effects). These criteria define the standards and benchmark by which missions' achievements, processes and practices should be assessed.

The paper then delves into the **“mission theory of action”, a tool specifically developed to track the specific mission processes and the systemic effects they are expected to generate to tackle a specific challenge**. It presents as an example a “theory of action” of missions that aim to support governments in achieving their net-zero targets .

Finally, the paper presents **a complementary tool to monitor mission-readiness levels**, i.e. how the mission's level of maturity, using the mission “ideal type” as a flexible benchmark. The tool, based on the OECD “MOIP design principles” framework, allows the different mission communities to self-assess their achievements, compare them with missions from other communities, and use these results to discuss necessary actions and possible ways forward. The tool was successfully tested in the context of Austria's efforts to mobilise relevant communities of actors around the five EU Missions.

These recommendations, criteria and tools for mission M&E have already been applied in several experiments. They will require more thinking and interactions, and will be applied and stress-tested in different contexts, at different levels. The OECD Mission Action Lab is currently in discussions with several missions to improve and apply these frameworks and tools in the near future. This is the condition for delivering effective and practical tools and methods capable of assessing what a French mission policy maker has called the *“supplément d'âme des missions”*.¹

1 Introduction

An increasing number of countries have adopted a mission-oriented innovation policy (MOIP) approach under different (often experimental) manners in the last five years to address complex systemic challenges (see definition in Box 1). Monitoring and evaluation (M&E) of mission-oriented innovation policy (MOIP) is critical not only to the success of each mission initiative, but also to the adoption and development of this policy approach – and, by extension, a range of other proactive policies aiming to transform sociotechnical systems.

M&E should help governments and their partners improve evidence-informed decision-making about the design and implementation of these new public policies – which in turn, will lead to more rapid and/or higher economic and societal impacts. More generally, it will promote learning within each initiative (from one stage of implementation to the next) and between initiatives (mutual learning). Finally, as for any other public policy, M&E will promote governmental accountability for the significant efforts devoted to implementing these policy initiatives.

M&E of MOIPs is also particularly important to justify the added value of this policy approach in already crowded and complex national governance and policy landscapes. This new policy approach raised significant expectations and attracted considerable political attention in the second half of the 2010s, not least after the launch of the EU Missions advocated by the political economist Mariana Mazzucato (Mazzucato, 2018 and 2019). This high visibility now translates into a strong demand for evidence of achievements and successes, and a strong imperative for missions to “prove themselves”.

Missions² are often seen as part of a broader trend of a return to proactive industrial policies (Juhász, Lane and Rodrik, 2023), notably to support the green and digital transitions. Against this backdrop, missions are also very much under scrutiny from those politicians, policy makers and academics who, based on evidence and ideology, doubt that a purposive, strategically planned and integrated approach can be an effective policy option. This is increasingly the case in the current context of growing criticism of missions in particular, and policies aiming to address societal challenges in general, and a pervasive call to (re)prioritise a bold competitiveness and growth agenda. MOIPs are therefore embedded in and coloured by a broad and cyclical debate that goes far beyond their remit and is not directly linked to their real (proven) performance. The M&E of missions will take place – and their results will be closely scrutinised – in this new context.

The practice of mission M&E is in itself also challenging, even though – at first glance – missions should make policy evaluation easier. Arguably, for “ideal-type” missions, the success or failure of a mission should be self-evident because these policies are result-oriented, and their objectives are expected to be “specific, measurable, achievable, realistic and timebound (SMART)”.¹ Mazzucato and Dibb (2019) describe missions as “granular and specific in that you must be able to answer whether or not they have been achieved.” One could therefore expect that mission results are measurable, meaning “quantifiable with existing metrics, or achievements that are evidently yes or no” (ibid). Simply put, and leaving aside the potential measurement problems, either the net greenhouse gas (GHG) emissions targets are reached at a certain date or they are not; the rate of fatalities from brain cancer can be below, equal or above some mission targets.

However, the reality of missions most often does not conform to this logic. Many missions do not have clear mission statements and targets – at least at the start – and their objectives are often too long term to allow measuring outcomes in a time frame that satisfies both accountability and learning requirements (OECD, 2024a). Also, different types of missions can have different types of objectives and blurred boundaries, and may be composed of different nested levels (MOIP initiatives, missions, interventions within the mission portfolio) (OECD, 2024b and 2024c). These features make the causal-effect relationship non-linear, and therefore difficult to measure in a detailed and robust manner. Furthermore, many missions are still very much in an exploration and learning stage. A recent OECD study showed that 81% of missions aiming to reduce GHG emissions directly or indirectly were initiated between 2018 and 2021 (OECD, 2023a). Since it often takes one or two years before missions start implementing activities, most missions currently have only a few years of concrete implementation and have not yet produced any tangible impact beyond their own structuring. Another important implication of this is that these “young missions” evolve in scope and shape, gradually forming the structures, processes and rules that will determine what they are, who is participating and how to achieve their objectives.

All these features which are the essence of missions complicate their evaluation and require specific efforts. This difficulty is compounded by the wickedness of the challenges they address, particularly their complexity and long time span (Baarslag, Geurts and Van Der Zee, 2024). However, although mission M&E is increasingly drawing attention and mobilising resources, it remains largely “uncharted territory” (OECD, 2021). Various rationales explain this situation, beginning with the complexity of these policies. The difficulties associated with their design monopolises most public resources and attention at the expense of other matters, including M&E. Other reasons might lie in some policy makers’ risk aversion, which hinders the development and application of new evaluation methods (Arnold et al., 2018). More practically, there was little time and attention available to think about M&E during the first years of the design and implementation of most MOIPs. These were often designed in a context of urgency and political pressure stemming from the need to respond to societal challenges, chiefly global warming.

So far, the few M&E activities around missions have used traditional approaches that are ill-suited for evaluating them (Rohracher, Coenen and Kordas, 2023). While the need for a systemic approach to public policy evaluation is well acknowledged (OECD, 2022), established methodologies and practices fall short of capturing the comprehensive benefits these missions are expected to offer at the system level, stemming from the integration of diverse knowledge, skills, interventions and funding streams across sectors. Capturing these “systemic effects”, which is critical for being fair to missions at a critical moment of the life cycle of this policy approach, requires developing new processes, methodologies and tools (Wise et al., 2022). The most promising initiatives in this respect come from academic work, but remain rather conceptual and have yet to be applied in real evaluation cases (Elzinga et al., 2023; Janssen et al., 2022; Wittmann et al., 2022; Haddad and Bergek, 2023; Molas-Gallart et al. 2021) or do not use a formal evaluation framework (Janssen, 2020; Haddad et al. 2023; Roth et al., 2021).

Against this backdrop, this paper first identifies the specificities of MOIPs, and how these determine the main features of a suitable approach to monitoring and evaluating them. Building on these broad framing elements, it then draws concrete implications in terms of evaluating the systemic effects and developing the corresponding evaluation criteria. The paper also proposes two specific tools respectively for the developmental evaluation of missions (the mission “theory of action”) and iterative monitoring of missions (the “mission-readiness monitoring tool”).

The work was carried out using a desk review, approximately 30 interviews and various interactions with mission practitioners and experts. A meeting organised at the OECD with selected mission-evaluation experts was a key milestone in the project (see Annex A). This paper also draws input from a study commissioned by the OECD Mission Action Lab in 2022 to an evaluation expert³ to examine a range of identified cases of MOIP evaluation (Arnold, 2022).

Box 1.1. OECD definition and typology of missions

OECD definition of MOIPs

One of the first task of OECD when it started working on missions was to establish a consensual definition of this new policy approach based on discussions within the CSTP Steering Group on MOIPs and through interactions with academics. As a result, the OECD defines MOIPs as a co-ordinated package of policy and regulatory measures tailored specifically to mobilise STI to address well-defined objectives related to a societal challenge, within a defined time frame. These measures possibly span different stages of the innovation cycle, from research to demonstration and market deployment; mix supply-push and demand-pull instruments; and cut across various policy fields, sectors and disciplines.

More succinctly, missions are defined as a space for collective action to address complex and systemic challenges. Collective action is the common thread that cut across the three building blocks of MOIPs: Strategic orientation: an MOIP involves a collectively developed agenda; Policy coordination: a dedicated structure of governance to take and monitor the effects of common or mutually consistent decisions; Policy implementation: a deliberately designed “joined-up” set of interventions. What is unique about MOIPs is not the presence of any of these building blocks taken in isolation – they are found in all STI systems – but rather their integration into a common organisational and institutional framework. Neither a whole-of-government national strategy for green hydrogen, nor a high-level cross ministerial committee for industry decarbonisation or a large-scale cooperative programme for a circular economy are sufficient to form an MOIP.

The MOIP building block

Although they can differ significantly in their design and types of objectives, MOIPs revolve around three building blocks

- Strategic orientation: They are oriented toward the achievement of clear objectives, formalised in terms of expected results (not inputs invested or actions to be implemented) and developed collectively.
- Policy co-ordination: They align the plans of various government actors in charge of different components (e.g., knowledge, technologies, funding, instruments pertaining to different policy fields, regulations) that are essential for achieving the collectively agreed-upon objectives.
- Policy implementation: They are implemented via a comprehensive mix of policy interventions and various initiatives to support a range of activities, from research to market launch and the formation of the required skills. These packages of interventions are deliberately designed to achieve the policy’s objectives. They can include joint implementation initiatives (from joint calls for proposal to cooperative, co-funded, schemes and programmes) that bring together multiple partners across policies fields and sectors.

Collective action lies at the core of each of these building blocks: an MOIP involves a collectively developed agenda; a dedicated structure of governance to take and monitor the effects of common or mutually consistent decisions; and, finally, a deliberately designed “joined-up” policy mix.

The key challenge of assessing missions consists in identifying and grasping the added value of this holistic dimension in missions, from the definition of the missions to their implantation. This is essential for i) the systematic collection of data that help mission managers understand progress in execution and performance in real time (monitoring of missions); ii) determine judgments about various criteria relative to how they produce their intended effects and the nature of these effects (see section 3 on mission evaluation criteria)

Source: Larrue (2021).

2 The characteristics of missions and their implications for evaluating and monitoring them

Developing mission monitoring and evaluation (M&E) processes and methods requires factoring in their complex, multidimensional and multilevel, and evolving nature. Understanding these specificities provides essential insights into developing suitable evaluation processes, methodologies and tools. This section reviews the main characteristics of missions and draw their implications for mission M&E.

Developing an evaluation framework that can be applied to different types of mission-oriented innovation policy (MOIPs)

By definition, all MOIPs share a similar approach, which consists in setting up a tailor-made institutional space for collective action towards resolving a societal challenge. However, this common policy approach comprises a variety of types which can differ significantly not only in scale and scope, but also in terms of objectives, type of governance, level of integration, etc.

Larrue (2021) has identified three main MOIP types: overarching mission-oriented strategic frameworks, challenge-based programmes and schemes, and ecosystem-based missions. Their main characteristics have significant implication for their M&E. **Overarching mission-oriented strategic frameworks** are the typical national missions that intend to mobilise a large part of a national innovation system towards a specific challenge (climate change, cancer, etc.). These broad, systemic and open-ended missions often resemble large, loosely co-ordinated and significantly decentralised umbrella programmes. These missions gather different authorities around shared goals, and it is often difficult to identify which interventions and activities concretely fall within the scope of the mission. The main difficulty in evaluating these entities is how to delineate these missions and assess their actual influence on the policies they include, especially those that are more remote from the organisations leading the mission.

Challenge-based programmes and schemes might be easier to evaluate at first glance: they are more limited in scale and scope, their objectives are clearer and more tangible (often of a technological nature), and they are most often project-based. Measuring their effectiveness, however, is difficult owing to the high risk of failure of the projects they support: a high level of failure of projects might not be an indicator of success at the overall programme level. In this case, it will be essential for evaluators to consider achievements at the level of the entire portfolio of projects. The impact of these technology-focused programmes is also difficult to assess, since diffusion and deployment are generally not included in their scope, while they are essential to MOIPs. While not specific to missions, this is a methodological challenge with which evaluators are still struggling.

Finally, **ecosystem-based missions** are initiatives that promote participatory approaches to mobilise large groups of actors around a common challenge. The ecosystems are empowered to define and update their

own strategic agenda regularly, with the government playing the role of broker and facilitator. An important issue is that linkages to policy interventions are weaker and more indirect in such initiatives. The public authorities leading these missions can limit themselves to funding the ecosystem orchestration, with the bulk of the funding for implementing the activities needing to be found through other policy instruments. This makes it more difficult to assess the impact of these initiatives, beyond ecosystem mobilisation and orchestration.

Overall, this disparity of MOIP types calls for adapting the evaluation questions and criteria to the type of MOIP, differentiating for instance between a challenge-based scheme or an overarching systemic mission. However, referring to the broad MOIP types presented above is not sufficient to adapt the evaluation to the specific mission at stake. Nor is it possible to rely exclusively on the mission objectives to understand what effects were expected and simply evaluate whether these were actually delivered. A key initial task for evaluators is to set the baseline to reconstruct the initial ambitions and how the different partners aimed to fulfil these through specific mission design, governance and processes. Mission evaluation needs to go beyond the “what” and explore the “how”, since it is the “how” that justifies the existence of these initiatives created to add directionality and co-ordinate existing policy interventions. Furthermore, in the early stages of MOIPs, evaluating the “how” is more informative and important to the long-term success of the mission.

A theory of action that accounts for the systemic effects would be instrumental in achieving this. If not made available by the mission itself, it should be developed as part of the evaluation, with the evaluation questions and criteria clearly linked to this theory of action (see below).

Accounting for the complex scope of MOIPs

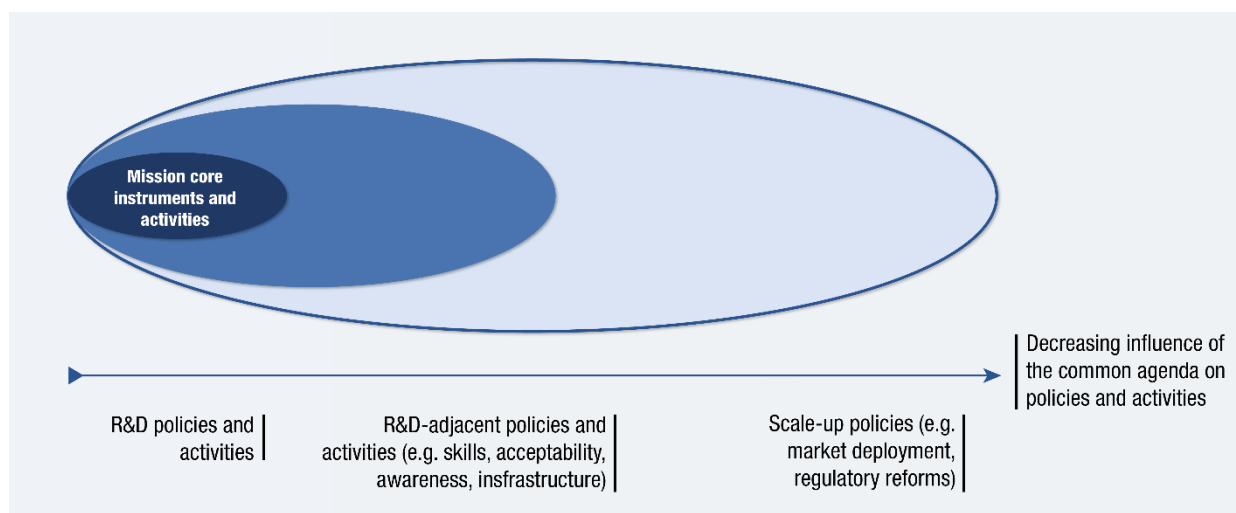
While some missions (notably the challenge-based programmes) have a well-defined scope, this is rarely the case for the wider, more systemic and less integrated missions (most often the overarching mission-oriented strategic frameworks and the ecosystem-based missions).

The mission boundaries are, therefore, often not clear-cut. The formal mission documents (plan, agenda, annual report, etc.) should in theory help to delineate the “mission space”, but these documents – especially the forward-looking agenda and plans – are more often than not a description of “what should be” than “what is” or “how to get there” in terms of the contribution of different actors and their respective interventions within the missions. Similarly, the documents used for broader communication to outsiders often tend to overemphasise the scope of the mission and the degree of co-operation within it, for obvious reasons of legitimacy and self-promotion.

The mission scope relates to one of the most critical processes – i.e. how missions enlist new actors, interventions, competencies, budgets and portfolios of projects and activities, beyond the initial coalition of committed partners who started the initiative – that largely determine their added value, hence the necessity to feature this aspect prominently in their evaluation. Previous OECD work has shown that many missions strive to go beyond the traditional science, technology and innovation (STI) policy actors and “attract” sectoral and regulatory authorities in charge of other interventions and activities (not least those related to deployment at scale, including through regulations, industrial policy measures and economic incentives).

In this view, missions are often better understood as composed of concentric circles (Figure 2.1), comprising a core which includes the activities that are directly governed by the mission (mostly those in the STI portfolio), and wider circles where policy instruments and activities are less influenced by the mission’s common strategic agenda (mostly activities in sectoral policy areas). At the very periphery of the mission lie the activities that are relevant to the mission’s challenge but are not connected to it in practice, i.e. they have been initiated, programmed and implemented in isolation from the mission decision-making processes.

Figure 2.1. Schematic map of a mission and degree of mission influence



Source: OECD

In most cases, delineating the mission scope will therefore be (at least to some extent) a matter of the evaluators' appreciation. Moreover, it should be a key dimension of the assessment itself, rather than a simple parameter of the evaluation. In light of the collected evidence, evaluators need to assess the sphere of influence of the mission along the three main dimensions of MOIPs: what interventions, projects and activities are steered by the mission's shared goals (mission orientation), and are mutually aligned towards these goals (mission co-ordination) and combined to contribute jointly to these goals (mission implementation)?

One key task for evaluators should be to "map" the missions and define as precisely as possible "what it is in and what is out" in terms of programmes, projects and activities. To this end, it is useful to distinguish the mission's formal scope and the mission's sphere of influence:

- The evaluator should identify the mission scope, i.e. who (and to what extent) were the partners engaged in the mission design (for example, the development of the strategic agenda) and implementation (the actors having committed policy instruments and resources to fulfil the mission's goal). This can be traditionally done using budget data, governance body information (including meeting attendance lists and activity reporting) and interviews.
- The sphere of influence is more difficult to assess and involves a higher degree of appreciation from the evaluator: how far does the influence of the mission extend? Has the mission overpowered the traditional vertical (i.e. silo) dynamics that predominantly steer interventions? Is the mission the strategic framework of reference for mission partners, or do they still act primarily towards the specific objectives of their own organisation in this area? Put simply, what is the extent and intensity of collective active action towards achieving the mission?

This mapping should consider the various layers of mission activities, as presented in Figure 2.1: the core activities, which are directly decided and funded "as part of the mission", within the sphere of control of the mission partners; and the other activities, which are less influenced by the mission or totally outside its realm, but contribute significantly to the achievement of the mission's goals.

Combining the evaluation of the different nested MOIP entities

MOIPs are multilevel, with different nested entities: individual projects are combined in portfolios, which are funded by different instruments gathered in a specific policy mix. They serve a specific mission which is part of a MOIP initiative, together with other “sister” missions. All these entities are subject to measures, from fully fledged evaluations to lighter monitoring and simple reporting. Since they operate at different levels at with different expected effects, their evaluation should have different foci (table 1).

Table 2.1. Evaluation focus at different MOIP levels

Evaluated entity/level	Specific evaluation focus
MOIP initiative	Achievement of the overarching objectives of the initiative Additionality of the mission-oriented policy approach (compared to more traditional approaches) Synergies and interactions between missions Timeliness and consistency with the overall strategic and policy landscape
Mission	Achievement of the specific mission goals Systemic effects of the mission Structures, processes and practices for maximising the systemic effects (including directionality, governance and portfolio alignment).
Individual policy interventions	Effectiveness, efficiency and relevance of each intervention relative to its specific objectives Consistency with missions
Individual projects and activities	Effectiveness, efficiency and relevance of each funded project and activity relative to its specific objectives Consistency with mission portfolio

Source: OECD.

Given its nested nature, the success of a MOIP initiative will depend on the achievements of its different entities. For instance, a given project’s impact depends in part on its embeddedness in a portfolio of other projects designed by the mission; in turn, the mission will depend on the ability of individual projects and activities to achieve their specific objectives and combine their respective outputs. Against this backdrop, there exists a need to combine, co-ordinate and balance the evaluations of the different nested entities at different levels.

Logically, the combination of the evaluations of the different entities participating in a MOIP should obey three principles:

- “Capillarity”: evaluations should build from the bottom up, with the results of each evaluation at one level informing the evaluations of the upper levels. The evaluations of the missions themselves are carried out as part of the evaluation of the MOIP initiative and inform the latter.
- “Subsidiarity”: evaluation at each level should focus on effects with maximum added value which cannot be achieved at a lower level. For instance, the evaluation of a mission should focus on the systemic effects resulting from the combination of various interventions, projects and activities. Such systemic effects cannot be captured at the level of any of these interventions, projects and activities in isolation.
- “Consistency”: the different focuses of evaluations at each level should be complemented by considerations of their immediate lower and upper levels to ensure overall consistency. For instance, when evaluating individual policy interventions, attention should also be paid to their degree of alignment with the mission in which they are embedded. Similarly, the evaluation should

consider each project's fitness in the portfolio. As the mission approach develops and becomes more mainstream, missions cannot be expected to simply co-ordinate existing and unchanged policy instruments and gather “passive” projects in their portfolio. Policy instruments will have to be modified (or new instruments created) to fit better within missions and match each mission's distinctive features (directionality, time frame, broad governance, etc.). Projects will be selected from the outset from the perspective of their possible added value as part of a larger and consistent portfolio.

Assessing the double additionality of missions

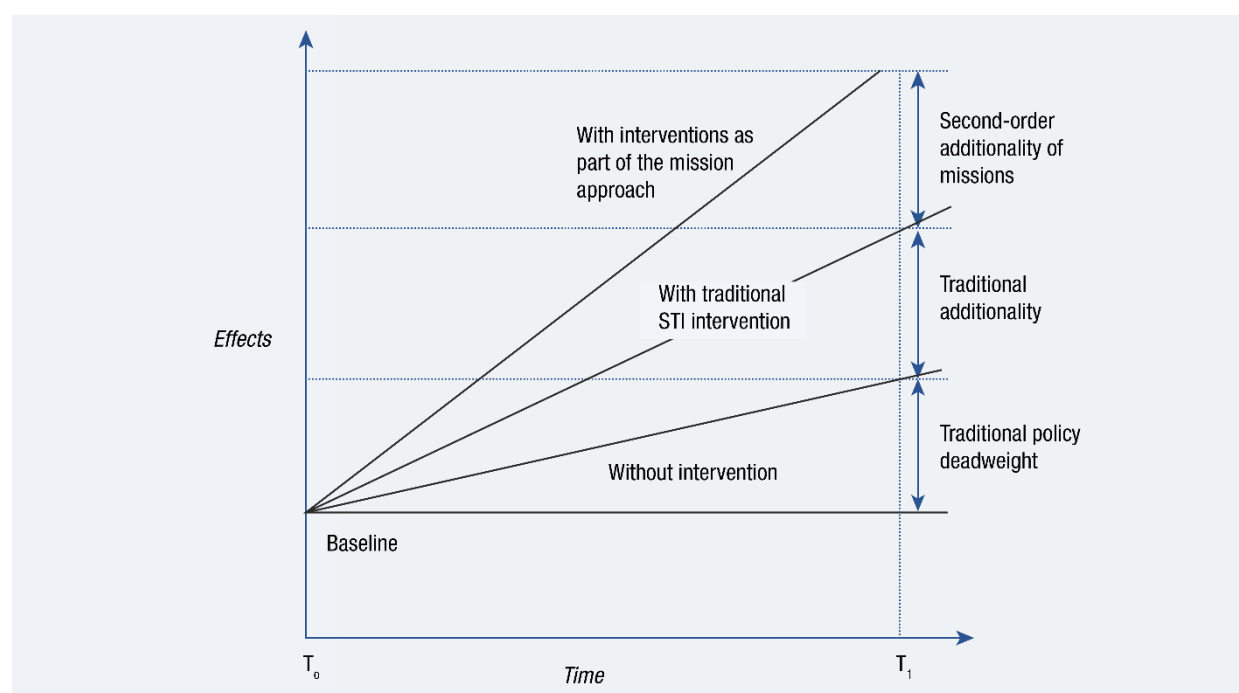
Additionality is essential in public policy evaluation. It measures what would not have been invested (input additionality), produced (output additionality) or performed by actors (behavioural additionality) without a government intervention. Although it most often cannot be measured precisely in the case of innovation policy because it involves information and counterfactual data that are barely accessible in most cases,⁴ it is important at least to structure evaluators' fieldwork and analysis.

The multidimensional and multilevel nature of MOIPs makes additionality even more difficult to discern. It involves assessing the additional costs and effects of adopting this particular policy approach relative to what would have happened if a more traditional STI policy had been used. Since missions build upon and integrate a package of existing interventions, the baseline (counterfactual) scenario is not the absence of intervention but the effects of these interventions without a MOIP. Mission evaluations should therefore assess whether the produced effects are superior to the mere addition of the effects of the combined policy interventions in their policy mix.

This is one of the most important challenges in evaluating missions. As an additional layer of co-ordination on top of existing instruments (and, in rare cases, new instruments specific to the mission), mission evaluation should provide a reliable assessment of “what that better coordination will yield for a better functioning system compared to a world in which that coordination would not take place”, which is “far more complex and complicated than evaluating the effect of a single instrument focused more or less directly on influencing the behaviour of individual entities” (Ter Weel et al., 2022).

Missions are, therefore, confronted with a “double-additionality” challenge (Figure 2):

- The “first-order” (traditional) additionality relates to the additional net effects of any given intervention included in the mission relative to what would have been the case in the absence of this particular public intervention (a “no-intervention scenario”). This additionality is in principle measured at the level of interventions (see above).
- The “second-order” additionality is specific to missions. It captures the additional effects of the whole set of interventions taken together as part of the mission, compared to a “no mission/policy-as-usual scenario”. This additionality should be compared to the additional transaction costs of missions resulting from the mission's (expected) stronger and broader co-ordination efforts.

Figure 2.2. The double additionality of MOIPs

Source: OECD (2023a).

The additionality of missions includes the additionality of each intervention it encompasses (first-order additionality) and the additionality of the bundling of these interventions within a mission (second-order additionality). Assuming that the first-order additionality is being taken care of at the level of the evaluation of individual policies, the evaluation of missions should focus on their “second-order additionality”. While no quasi-experimental methods can be used to that effect (even less so than for traditional STI policies), evaluators mainly rely on qualitative evidence drawn from desk reviews and interviews with mission leaders and partners. The evaluation questions underpinning these fieldwork activities should be designed to capture additional (systemic) effects: are the objectives clearer than in other, more traditional schemes/programmes in relevant areas? Are the objectives more ambitious? Has the mission accelerated the development of new solutions or allowed developing solutions (e.g. with a higher performance or wider reach) that would have been out of reach of transitional interventions? The evaluation questions should also consider the possible “negative additional effects” of the mission (displacements, crowding out, lock-in, etc.). For instance, has the mission missed the emergence of new areas because its funding is overly targeted? The evaluation should also take into account a potential increase in the level of administrative complexity, rigidity and costs owing to the additional co-ordination requirements.

Embracing the evolutive nature of MOIPs

Many missions start with broad objectives, which are gradually refined by the partners as negotiations progress, trust grows, and additional information is provided and analysed. Once they are set, objectives can still be revised to reflect not only changes in context and strategy, but also the interests of new actors who come on board to complement the initial partners. This is particularly true for ecosystem-based mission approaches, since the partners themselves define and revise the strategic agenda regularly. In these types of initiatives, the objectives and the strategic agenda (or roadmap) in which they are embedded

are not the starting point, but rather an important result of the mission. This leads to a “moving target” issue for evaluators, who need baseline objectives against which to assess the mission achievements.

Furthermore, not only the objectives but the mission itself is evolving. Changes are particularly significant in their three main building blocks (see Box 1) during their first years, when they gradually negotiate their objectives and scope (strategic orientation), shape their governance and composition (policy coordination), and establish corresponding processes and, possibly, joint implementation practices and instruments (policy implementation). However, missions keep on evolving continuously towards their ‘desired state’ throughout their life cycle. Following an iterative design process, missions may start with design features and processes that are far from the mission ‘ideal type’ and incrementally improve, increasing what might be called their ‘mission-readiness level’.

In a pattern that has been long studied in evolutionary studies, the mission as an institution co-evolves with the solution it is supposed to generate (e.g. Saviotti, 2005). In this case, the evaluators cannot assume consistency in the mission design and implementation, as it evolves over its life.

This feature does not lend itself easily to traditional evaluation approaches – especially of the summative type – that consider mainly the baseline and the endline. Following a developmental evaluation approach, an important part of the assessment should be the extent to which “the mission becomes a (‘real’) mission” as it deploys its activities and learns. The evaluation should consider not only the effects of the mission, but also i) how it evolves in terms of design, governance and processes (objective-setting, funding, portfolio management, etc.); and ii) whether these evolutions correspond to what was expected and motivated the adoption of a mission approach, and the reasons for any observed gap.

This process dimension of the evaluation in the context of mission yields a broader understanding of behavioural additionality. The success of the mission will depend on the policy makers and their partners (i.e. not only the beneficiaries) changing their daily practices, routines and mindsets. In a sense, in missions, policy makers should be considered as a policy target group of their own, since important behavioural changes are also expected on their side. Evaluation could therefore also cover questions such as, has the mission become the framework of reference for strategic thinking for the actors involved? Does the mission overcome vertical thinking and practice? Is it a small “experimental bubble” within organisations, or has it become more central and pervasive?

This evolving nature of missions calls for a developmental approach where the evaluators are engaged in the intervention to be evaluated on a more continuous and long-term basis. The evaluation is ongoing, rather than one-off or carried out at specific points in time. It provides continuous input through feedback loops, from the evaluation to the intervention design and implementation. The evaluation should serve the mission’s progression, clearly identifying the gaps and pointing towards needed improvements, without discouraging efforts and hindering learning processes.

Another implication is the need to “stage” different types of evaluations with different expectations. Early in the mission’s life, the focus should be on the formative/developmental evaluation of the mission approach (i.e. how the mission processes evolve to produce the systemic effects). As the mission unfolds, a summative evaluation of the mission (i.e. how it progresses along its milestones towards the realisation of the final objectives) can complement these evaluations.

Adopting a nuanced approach: More is not always better

While a mission’s evaluation should be underpinned by an ambitious ideal type to assess its mission-readiness level (i.e. its progression towards a more “fully fledged” mission), as mentioned above, the precise evaluation benchmark needs to be adapted to each specific mission case. Not all missions should aim for the highest position in all three mission dimensions:

- Regarding orientation, some missions might consider a less directional approach (at least at the outset) to give mission partners more autonomy to set their strategic agenda (this is particularly the case for ecosystem-based missions).
- Regarding co-ordination, there exist different types and intensities of co-ordination, and the broadest and strongest form of co-ordination is not adequate for all missions.⁵ The need for co-ordination will largely depend on the national institutional specificities and the characteristics of the challenge (level of uncertainty, complexity, contestability). Excessive co-ordination can generate heightened transaction costs, complexity, rigidity, slowdown of action, lack of leadership, dilution of priorities and blurred accountability. Experiences in some of the MOIPs that have deployed a fully systemic approach, such as the Dutch Top Sectors and Mission-Driven Policy, shows that complex co-ordination can generate significant “mission fatigue”.
- Similarly, when it comes to mission implementation, full integration of the different instruments is not always either possible or desirable. In some cases, a smart combination of interventions can be sufficient, less costly and more flexible.

These three arguments call for a more nuanced and strategic approach, grounded in an ex-ante assessment of interdependencies and the need for information exchange, plan alignment and joint actions. It is important to consider the structural features of the challenge area being addressed by the mission. The level of complexity, uncertainty and contestability, in particular, will have different implications in terms of the appropriate design of the missions. While it is not possible to propose a deterministic view which holds that an optimal design exists for any given challenge, understanding how the characteristics of an area have influenced the mission design and challenges should be one task of the evaluators.

3 A tailor-made suite of mission-evaluation criteria, processes and tools

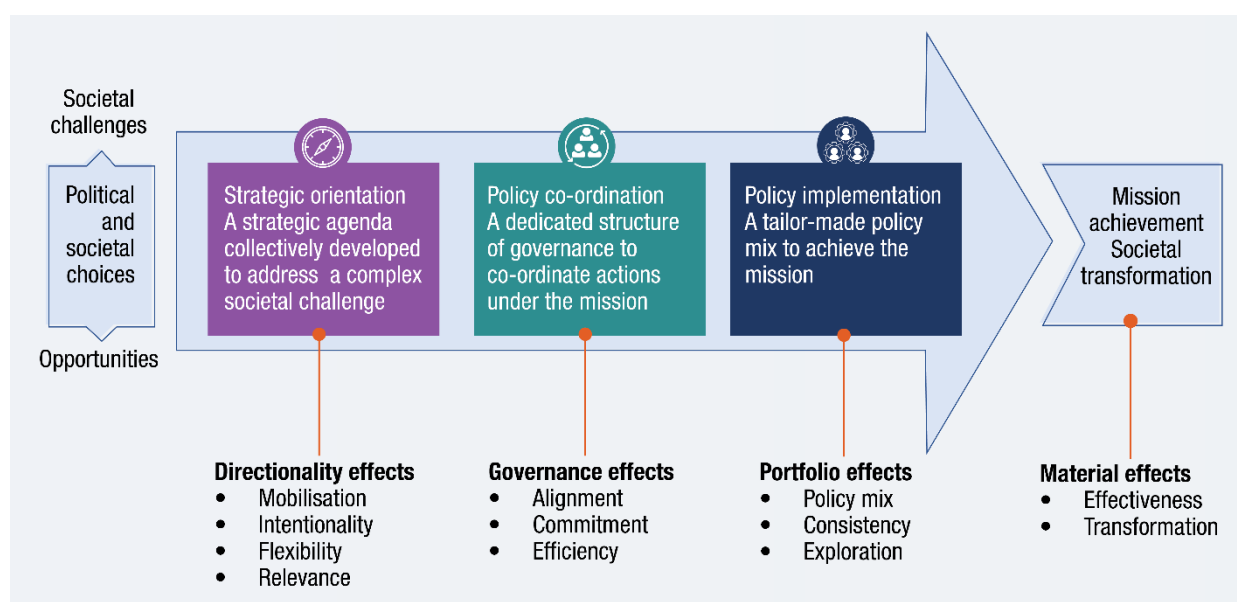
The previous section focused on the specific features of mission-oriented innovation policies (MOIPs) and their implications for monitoring and evaluation (M&E). Notably, it points towards the use of a developmental evaluation approach that focuses on tracking the evolution of the mission design and processes, and assessing their systemic effects. This assessment of the mission additionality – which firmly establishes its *raison d'être* – requires a good understanding of the unique “theory of action” that underpinned the chosen mission design and processes in a particular institutional and thematic context. This section proposes mission-evaluation criteria, processes and tools that are consistent with this evaluation approach.

Setting mission-evaluation criteria

As mentioned in Section 2, MOIP evaluations should focus on assessing the specific “second-order additionality” of this policy approach, meaning the extent to which missions generate systemic effects (positive or negative, intentional or not) on top of those produced by the instruments they intend to steer, co-ordinate and integrate. Missions’ expected systemic effects – i.e. the effects produced by the collective actions within missions (collective development of a strategic agenda, alignment of plans toward these objectives, bundling of instruments and portfolio management). These effects are the reason for this policy approach, which addresses challenges for which traditional (i.e. less directional, co-ordinated and integrated) innovation policy approaches are considered less effective. Since missions are by nature tailored to a specific challenge in a particular context (subnational, national, international), no two missions are alike in principle, and the evaluation will need to be adapted to each mission. However, there exist archetypal effects that are expected from any mission, in different proportions and at different degrees of intensity, depending on the mission objectives and context.

Based on previous work and fieldwork conducted as part of this study, the OECD proposes four main groups of mission effects:⁶ directionality effects, governance effects, portfolio effects and material effects. For each of these, specific criteria can be derived to guide the evaluative work. These criteria define the standards and benchmark by which missions’ accomplishments or suitability should be assessed (Figure 3). These criteria are complementary to traditional evaluation criteria, such as those of the OECD Development Assistance Committee (DAC),⁷ which are used at the level of the evaluation of individual interventions.

Figure 3.1. Mission systemic effects and corresponding evaluation criteria



Source: OECD.

Directionality effects

Directionality effects refer to the strategic orientation and focus of public interventions targeting specific societal challenges. This involves setting clear, ambitious, measurable/evaluable and legitimate objectives to drive research and development (R&D) in a purposeful direction, often addressing complex, multidimensional issues. This contrasts with traditional neutral or functional STI policies, which focus on specific market failures to address predominantly growth- and competitiveness-related objectives.

The proposed criteria under this group of effects are:

- **Mobilisation:** the extent to which the mission brings together a set of public and private actors with the resources, skills and authority necessary to address a complex challenge (regardless of their belonging to specific institutions, sectors or disciplines) to develop and implement a common strategic agenda. The related effects proposed by evaluation experts during the Mission M&E workshop (see Annex A) were “inspirational” and “motivational” effects.
- **Intentionality:** the extent to which the mission has involved a wide array of public and private stakeholders (including citizens) in the collective development of clear and ambitious objectives with accompanying SMART, as well as a strategic agenda that spells out the different possible pathways to meet these objectives.
- **Flexibility:** a mission’s capacity to revise as regularly as necessary its objectives and strategic agenda, based on information derived from strategic policy intelligence and lessons learned from implementation.
- **Relevance:** the extent to which the strategic agenda is pertinent to the underlying problem and effectively addresses the underlying causal issues; the level of alignment of the envisioned actions with the scale and scope of the complex challenge.

Governance effects

Governance effects refer to the collaborative steering, resource mobilisation and allocation, management and monitoring of various public interventions aimed at the mission objectives to ensure that innovation efforts are aligned with the mission and are conducted efficiently responsibly, transparently and accountably.

The proposed criteria under this group of effects are:

- *Alignment*: the extent to which the investment and activity plans of the mission's public and private partners are coherent and consistent with the mission objectives
- *Commitment*: the extent to which the mission attracts resources (e.g. financial, technical, in-kind) from various public and private actors to implement the mission, regardless of their specific institutions, sectors and disciplines
- *Efficiency*: the extent to which the mission delivers or is likely to deliver results in an economic and timely way, and with a manageable level of complexity.

Portfolio effects

Portfolio effects describe the various synergies stemming from a combination of innovation projects and investments that were strategically selected and supported by different public interventions to collectively meet the mission objectives. This approach recognises that no single project or activity can solve multifaceted issues. It enhances the likelihood of fulfilling the mission by distributing risk, leveraging different types of complementarities (e.g. along different stages of the value chain, different generations of technologies, different components of a sociotechnical system), and promoting a holistic approach to innovation.

The proposed criteria under this group of effects are:

- *Policy-mix integration*: the extent to which the mission combines efficiently and effectively a variety of complementary policy interventions to achieve the mission
- *Portfolio consistency*: the extent to which the mission supports a coherent and aligned set of innovation projects and activities that collectively contribute to the mission's objectives
- *Exploration*: the extent to which the mission investigates a wide range of innovative (including unconventional) ideas, solutions and pathways and cultivates openness, creativity and experimentation in the pursuit of the mission's objectives.

Material effects

Material effects refer to the concrete outcomes and impacts of missions in relation to their objectives. As highlighted in Section 2, missions must not only fulfil their objectives, but also fulfil them more effectively and efficiently than would have occurred with traditional approaches – or they should fulfil objectives that are beyond the reach of these traditional approaches.

The proposed criteria under this group of effects are:

- *Effectiveness*: the extent to which the mission has achieved, or is expected to achieve, its objectives and meet its targets in the relevant time frame, including any differential results across groups or components of the mission
- *Transformation*: the extent to which the mission significantly alters existing technical, economic, social or environmental systems, practices and structures to meet its objectives. The contribution

to system transformation (possibly using a list of technological innovation-system functions – see Section 3), should be considered only when relevant to the mission’s objectives and/or when specifically asked by the evaluation commissioners.

Annex B presents some of the main questions associated with each criterion.

Setting the mission baseline: The “mission theory of action”

What is a theory of action?

Theories of change are increasingly used to guide the design, implementation and evaluation of complex and adaptative interventions. They are logical diagrams that display the expected processes of change on the material side, i.e. the objects upon which the policy interventions act. They link, for instance, the development of new carbon capture and storage systems to a reduction of GHG emissions through a succession of causal linkages.

Alongside these theories of change, theories of action are tools that track the specific mission processes and the systemic effects they are expected to generate to tackle a specific challenge. In a nutshell, a theory of action is inward-looking and process-oriented. It is “the delivery model for a Theory of Change” (George, 2024). It describes how the mission is designed and set up to achieve the mission. It maps the chains of causality that connect the way missions are designed and governed to the impact they are expected to produce. Rather than focusing only on the material dimension of missions (is the mission being achieved and the selected challenge solved?), mission theories of action track the key processes that lie at the heart of the MOIP approach, and allow assessing whether they unfold as expected and produce their “systemic effects” (i.e. is the adopted approach effective at tackling the selected challenge?)

Theories of change and theories of action are therefore two faces of the same coin, providing complementary views on the expected process of change that should lead to fulfilling the mission. Mission theories of action can arguably prove particularly instrumental to both mission managers and evaluators in a developmental approach. In the case of MOIPs, theories of change should surface the expected causal relationships between the problems these policies tackle and their desired goals, inputs, outputs, outcomes and impacts (Janssen et al., 2021; Hekkert et al., 2020). These maps of mission “contribution claims” can prove particularly instrumental in guiding and monitoring their portfolio of activities, to verify they are coherent and aim in the right direction.

An example of a mission theory of action

The OECD used a theory of action to assess whether and under which conditions MOIPs are better suited than traditional STI strategic and policy frameworks to supporting countries’ efforts to meet their net-zero target (OECD, 2024a, Figure 4). The OECD used this tool to analyse a database of about 100 “net-zero missions” and 20 in-depth case studies.

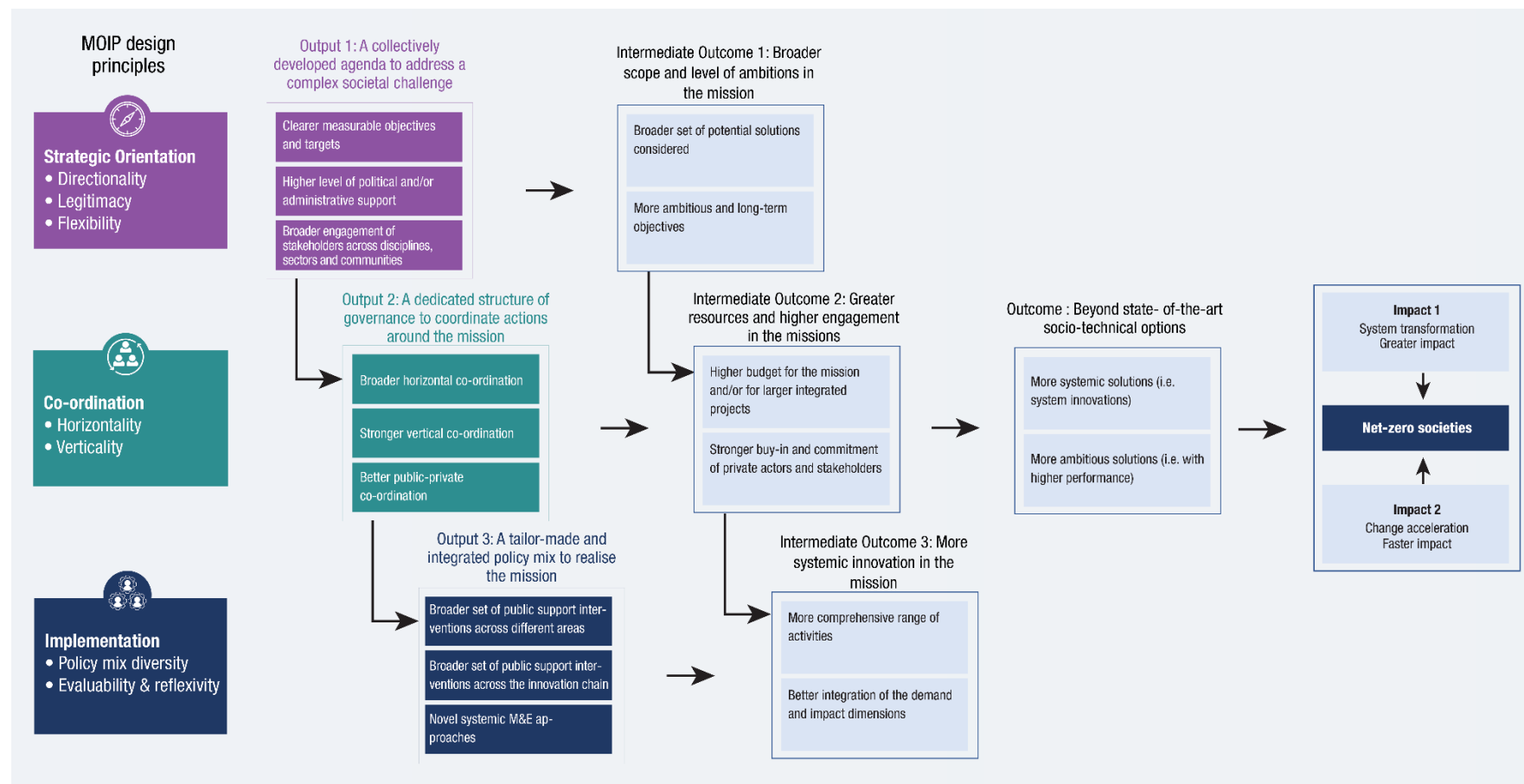
Like a mission theory of change, a theory of action consists of several stages, from the mission design principles to the achievement of the mission’s objectives:

- The starting point of the theory of action is the description of its main features along the three dimensions of orientation, co-ordination and implementation. These features describe the processes and structures (including governance bodies) that set objectives and targets, co-ordinate vertically and horizontally actions, build synergies between funding streams, combine instruments to support a consistent portfolio of activities, etc.

- The second stage of the theory of action features the *intermediate outcomes* of the MOIPs. For instance, it is expected that gathering multiple public and private actors from different disciplines, sectors and policy areas allows developing a shared agenda that has strong legitimacy among these actors and beyond, as well as other features, notably a high level of directionality and intentionality. In turn, this shared agenda is expected to include a broader set of potential options directed towards more ambitious and long-term objectives.
- In combination with the elements of the policy co-ordination building block, missions produce *outcomes*. A wide range of actors across all silos, who are strongly committed to achieving the mission by exploring a broader set of potential options and supported activities and benefiting from greater resources, could design more ambitious solutions that could be applied more systemically.
- At the end of the causal chain, the *impact* of a MOIP is the expected fulfilment of its mission.

This framework of outputs, intermediate outcomes, outcomes and impacts underpinned the structure of the interview guidelines.

Figure 3.2. Example of a generic mission theory of action in the net-zero area



Source: OECD (2024a).

For the purpose of the OECD study, this theory of action was voluntarily left rather generic to fit a very diverse set of missions. When used to design and/or evaluate a specific mission (or MOIP initiative), it specified its processes, mechanisms, structures of governance (with the expected composition and mandates), management teams, etc. When used to evaluate a mission, it provided a key tool to track the mission's gradual evolution towards a higher degree of "mission readiness" and expected systemic effects.

This type of theory-based evaluation will require a different role and type of "embeddedness" of both evaluators and academics. The evaluator will need to be closer to and evolve more continuously with the missions being evaluated. Like real-time evaluations, these evaluations are ongoing, carried out alongside the development of the mission and aim to provide continuous and regular feedback. This approach contrasts drastically with the "arm's-length" and punctual approach adopted by evaluators in summative and formative evaluations to ensure their independence. In some cases, the embeddedness of evaluators and academics in the missions, reflecting the theory-driven nature of these evaluations, will lead to action-research types of exercises. Furthermore, understanding the process side of the mission effects and how they are produced (or not) will require considering the latest developments in science policy studies, notably in relation to system change and transitions.

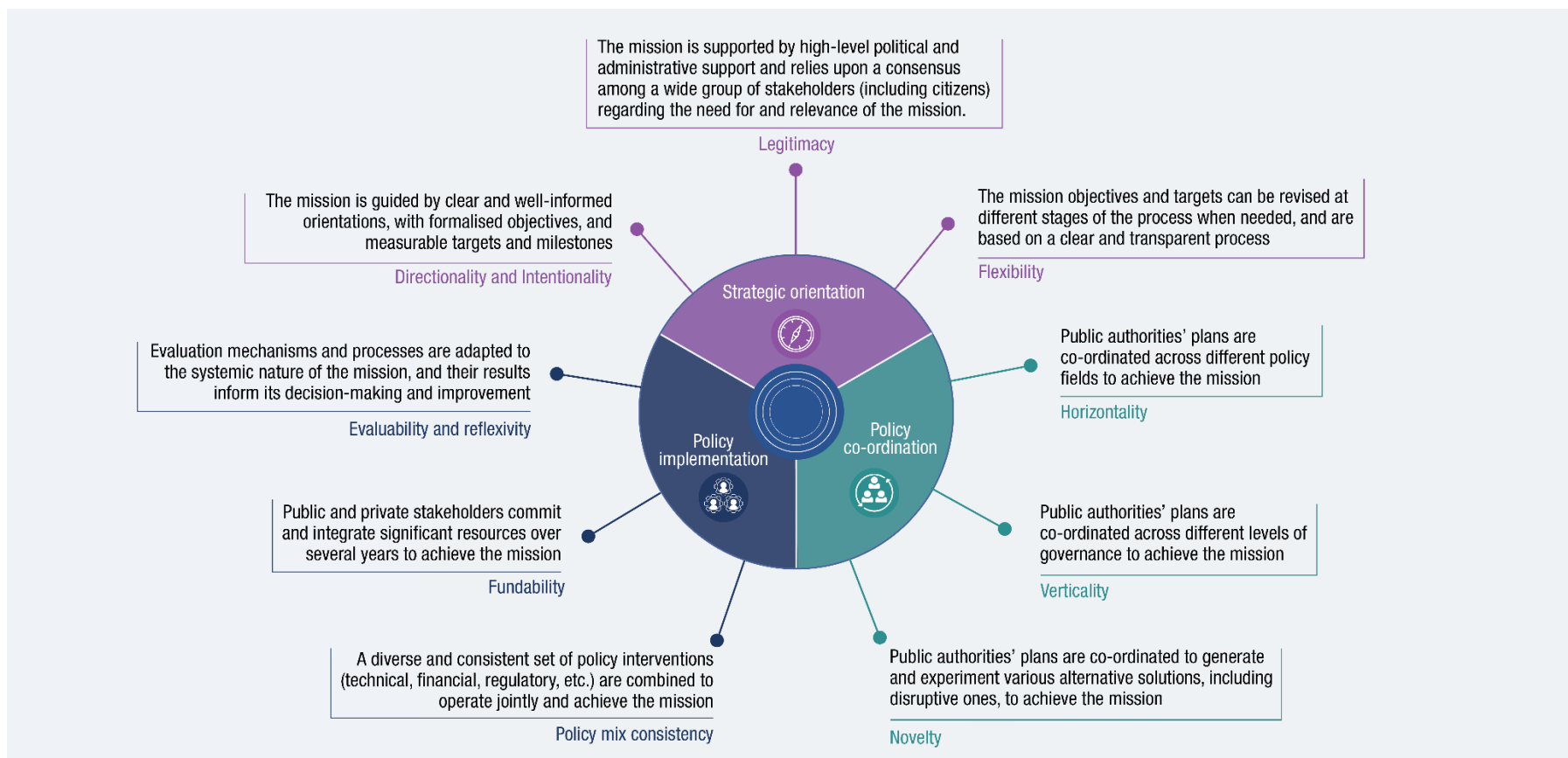
Monitoring the progression of the mission-readiness level

Missions continuously learn and evolve, not only in terms of their strategic agenda but also their composition, structures, resources and modes of action. As explained in Section 2, a key task in mission M&E is to track the process of the "mission becoming a mission" – i.e. increasing its mission-readiness level. This section proposes a tool to assess and monitor this process. It was developed by the OECD and tested in Austria, in co-operation with Austrian authorities.

The mission design principles as a map and a compass

This framework builds upon the OECD "MOIP design principles" (OECD, 2021). These principles define an ideal type of MOIP, while leaving sufficient flexibility to accommodate the institutional specificities and strategic vision of each country or community (Figure 5). Although few initiatives – if any – can "tick all boxes", this framework provides mission leaders with a useful and flexible benchmark.

Figure 3.3. Design principles of MOIPs



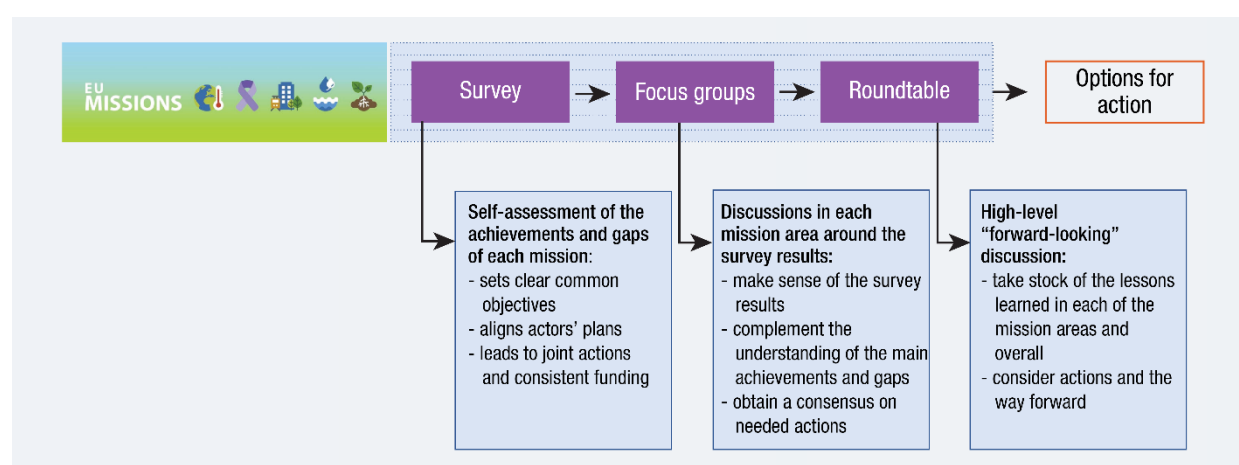
Source: OECD. By using these design principles to interrogate and compare missions, mission partners can derive:

- i. a map to assess their current position – a self-assessment of their progress in the mission design and implementation “journey”, using a framework developed specifically by the OECD to analyse missions
- ii. a compass to show the way forward.

A real-life test of the mission-readiness assessment tool

In co-operation with Austrian authorities, this analytical framework has been further developed into a mission-readiness assessment tool and tested in the context of Austria's efforts to mobilise relevant communities of stakeholders around the five EU missions.¹ The so-called “demonstrator” consisted of three main steps (Figure 6).

Figure 3.4. Main steps of the Austrian demonstrator mission-readiness level monitoring tool

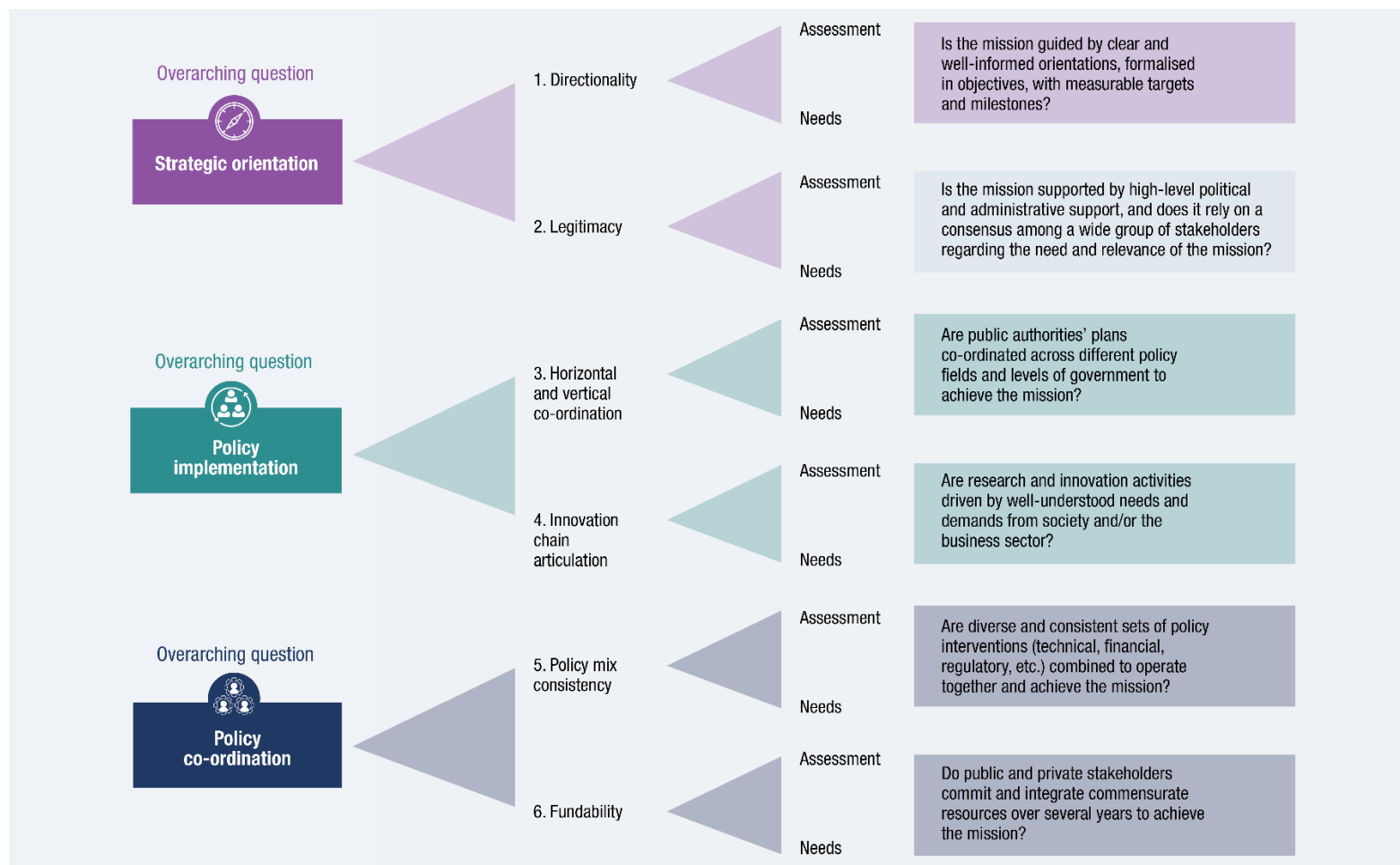


Source: OECD.

The first step consisted in an assessment survey sent to each mission community.² Each design principle was informed by a few questions (Figure 7). To better guide action, each question distinguished between two dimensions:

- assessment (how does the mission perform on the design principle?)
- needs (what would need to be improved?).

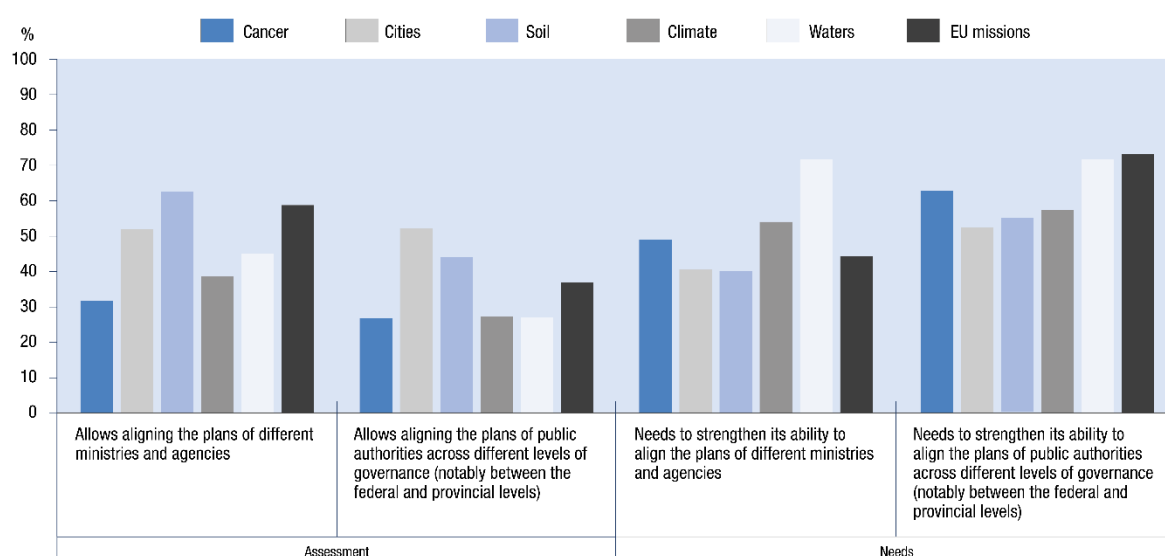
Figure 3.5. Generic structure of the mission-readiness level monitoring survey



Source: OECD.

The results lent themselves to various types of meaningful comparisons and were used as key materials for supporting participative focus groups and workshops. By way of illustration, Figure 8 presents the results of the five missions and overarching working group for the questions related to the horizontal and vertical co-ordination design principles.

Figure 3.6. Results of the questions informing the horizontal and vertical co-ordination design principles



Note: Share of survey respondents that agree or strongly agree with the following statements regarding the mission.

Source: OECD.

The comparisons of the readiness levels of different design principles for one mission, as well as the comparisons of the different missions across one design principle, revealed marked profiles that were very effective in supporting informed discussions during subsequent workshops in each mission area. A final all-missions roundtable complemented the process. The conclusions of this self-assessment exercise were synthesised in a final report, which included overarching and mission-specific recommendations (OECD, 2023b).

The tool will be refined and tested in different contexts. One particular line of improvement consists in developing relevant evaluation rubrics to define and grade the different levels of expected performance for each design principle.

It can be useful at all stages of the mission cycle to track the progress of the mission along the most crucial dimensions of mission orientation. This is particularly suited to assessing the need and capacity for a mission approach at the crucial milestone of the end of the preparatory stage, before the mission's launch.

4 Conclusion

Missions are living entities. Not only their objectives but also their forms, shapes and content evolve as they learn and gradually come closer to what they are expected to be and perform. Against this backdrop, mission M&E approaches are also changing and improving. However, this co-evolution only functions through real case-based studies and evaluations. These provide essential feedback on mission processes and effects, and are also key to refining the M&E approaches and methods.

Relying upon desk research, previous OECD work on missions, inputs from mission M&E experts (see Annex A), a dedicated benchmarking study and a Mission Action Lab experiment in Austria, this paper proposes a M&E framework combining different types of evaluations at different stages of a mission's life cycle, with tailor-made criteria and tools to capture the systemic effects it is expected to generate.

The results of this work, which has already been applied to several experiments, will require more thinking and interactions and will be “stress-tested” in other contexts, at different levels. The next steps will be as follows: i) apply the developmental evaluation framework and guidelines proposed in this paper to real cases to define practical evaluation terms of reference, including concrete indications on the scope, the steps, the criteria to be used, the related evaluation questions, the methods and the overall governance of the evaluation; ii) improve and apply the tools to other mission M&E cases in the near future (the OECD Mission Action Lab is currently in discussions with several missions for that purpose); and iii) define a dynamic approach to mission M&E, with clear indications regarding the types of M&E activities to be carried out at different moments of the mission life cycle.

Engaging in real cases of mission M&E is the condition for delivering effective and practical tools and methods that are able to assess what a French mission policy maker called the “supplément d'âme” of missions.

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Endnotes

¹ This could be translated as the “extra touch of soul of missions”.

² The terms “MOIPs” and “missions” are used interchangeably in this report. When referring specifically to missions within MOIPs, the terms “MOIP initiatives” and “individual missions” are used to avoid ambiguity.

³ Erik Arnold, Technopolis and University of Twente.

⁴ Quantitative assessments of additionality have often proven difficult due to the difficulty of running counterfactual scenarios through pairing techniques. While these have proved their added value in random control trials, notably in clinical, development aid or education programme evaluations, their use in the STI policy domain is more difficult. The beneficiaries of an innovation scheme are most often of a different nature than those who did not apply to the scheme or were not selected. The resulting selection bias makes any comparison between beneficiaries and non-beneficiaries (constituting the counterfactual sample) hardly meaningful.

⁵ Les Metcalfe (1994) proposed a scale of co-ordination that ranges from independent decision-making by ministries/agencies at the lowest level to government priorities that are collaboratively prepared and give “a definite pattern and direction to the work of ministries”. Building on this seminal work, a number of scholars have proposed different scales of co-ordination that allow a more fine-grained analysis.

⁶ These effects and corresponding criteria correspond to the level of missions specifically rather than MOIPs as a whole, since it is at the level of each mission that the bulk of additional effects are expected to be produced.

⁷ The DAC criteria are: relevance (is the intervention doing the right things?); coherence (how well does the intervention fit?); effectiveness (is the intervention achieving its objectives?); efficiency (how well are the resources being used?); impact (what difference does the intervention make?); and sustainability (will the benefits last?) (see OECD, 2021).

¹ The five EU Missions are: 1. Adaptation to Climate Change: support at least 150 European regions and communities to become climate resilient by 2030; 2. Cancer: working with Europe's Beating Cancer Plan to improve the lives of more than 3 million people by 2030 through prevention, cure and solutions to live longer and better; 3. Restore our Ocean and Waters by 2030; 4. 100 Climate-Neutral and Smart Cities by 2030; 5. A Soil Deal for Europe: 100 living labs and lighthouses to lead the transition towards healthy soils by 2030 (see https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe/eu-missions-horizon-europe_en).

² In Austria, five Mission Action Groups were formed corresponding to the five EU mission areas. This framework is led and monitored by a task force called the “EU Missions Working Group”.

Annex A. The Mission M&E expert meeting

An expert meeting, held on 3 and 4 October 2023, aimed to develop collectively some of the critical components of an M&E practical framework through prior document-based consultations and structured discussions. It gathered some key mission M&E experts (see the list below) for two half days. Several OECD officials from across the organisation (STI, Public Governance, Development Co-operation, Internal Evaluation) also attended the meeting. All of them have confronted the challenge of the mission M&E from different angles and in various contexts (theoretical and/or practical) related to regional, national or EU missions.

In stark contrast with most traditional workshops, where each expert presents their own work on a specific issue, this meeting proceeded through successive sessions that contributed to the final aim established above. The experts were asked to work in the broad context of the analytical framework developed by the OECD. An early part of the meeting focused on a critical review and revision of this framework to ensure it is relevant and does not miss important points. Once this was accomplished, experts worked within this framework to derive the practical resources (including the evaluation's scope, questions, criteria, methodological approaches and steps) needed to guide policy makers and evaluators as they attempt to monitor and evaluate missions. The discussions were expected to lead towards clear and practical guidance – i.e. away from participants' theoretical frameworks, while building on these.

Discussions were informed by the inputs provided by experts before the meeting via a dedicated template. The experts' inputs were compared, synthesised and used as the main background material for the meeting.

Both the template and meeting agenda were structured along missions' groups of (expected) effects. Having listed these groups of effects, experts were asked to focus on each of them successively to determine their content and definition; the relevant criteria, markers and indicators; and the methodology for assessing them in concrete-use cases. The template and meeting, therefore, used a logic of inquiry inspired by theory-based evaluation approaches, whereby the identification of the expected causal linkages that underpin missions' effects is a prerequisite for evaluation fieldwork.

The main questions discussed at the expert meeting were:

1. Does the OECD mission M&E framework seem adequate for underpinning M&E efforts for mission-oriented policies? What is missing or possibly misleading? How could its content and formulation be improved?
2. What are the two or three main groups of effects expected from typical mission-oriented policy initiatives?
3. For each group of effects:
 - What are the main mechanisms/causal linkages underpinning this type of effect?
 - At what level(s) (MOIP initiative, specific mission, specific instrument within the mission) should evaluation capture this type of effect?
 - What are the corresponding evaluation questions and evaluation criteria/markers/indicators?
 - How can one assess this type of effect? What are suitable tools/methods?

The results of the consultation and meetings were instrumental in developing the arguments and recommendations presented in this paper.

Workshop attendees

The following experts provided inputs through a template-based consultation and participated in in-person meeting at the OECD in Paris:

- Ptries Boekholt, Director, Innovation Policy Matters
- Emily Wise, CIRCLE, Lund University
- Stephanie Daimer, Co-ordinator of Business Unit Policy for Innovation and Transformation at ISI Fraunhofer
- Carolina Resende Haddad, OECD, formerly Department of Technology Management and Economics, Chalmers University
- Matthijs Janssen, Assistant professor at Copernicus, Utrecht University and Principal Scientist at Dialogic
- Michal Miedzinski, Economic and Policy researcher at JRC Sevilla and Senior Research Associate at the UCL Institute for Sustainable Resources (ISR)
- Rainer Kattel, Deputy Director and Professor of Innovation and Public Governance at IIPP, UCL

OECD officials:

- Benjamin Kumpf, DCD/RREDI
- Carlos Hinojosa, EVIA/IDE
- Caroline Haddad, STI/STP
- Charles McIvor, STI/STP
- David Jonason, GOV/INDIGO
- Olivia Sabini-Leite, GOV/INDIGO
- Philippe Larrue, STI/STP
- Piret Tõnurist, GOV/INDIGO

Annex B. Mission effects, evaluation criteria and examples of relevant evaluation questions

Mission effects	Evaluation criteria	Main evaluation questions
Directionality effects	Mobilisation	Is the mission enlisting the public and private actors necessary to solve the targeted challenge? Has the mission helped achieve consensus and increase buy-in/ownership of the initiative?
	Intentionality	Have the broad challenges and orientations been translated into clear, measurable and actionable objectives/targets, with a clear timeline? Has the mission allowed setting more ambitious/challenging objectives, and in what ways?
	Flexibility and reflexivity	Has the mission been able to adjust the mission objectives or targets as needed, without reducing the level of ambition? Has the mission benefited from effective information feedback loops from activities to question the relevance and feasibility of objectives? To what extent has the mission adapted its mechanisms and processes for critical self-assessment and responsiveness to emerging information, changing conditions, stakeholder feedback and M&E, in order to improve its capacity to fulfil its objectives?
	Relevance	To what extent does the mission address a clearly identified challenge? To what extent is a MOIP approach appropriate to address this challenge (compared to traditional/existing policies) How well does the program align and is connected to government strategic and policy agenda?
Governance effects	Alignment	Has the mission been able to connect and align the strategies and plans of different policy bodies across jurisdictions and levels?
	Commitment	Has the mission led to an increase and/or broadening of public and private resources invested to meet the mission objectives?
	Efficiency	How well are the resources dedicated to the mission being used? What has been the mission's effect on the level of transaction costs and complexity borne by mission partners?
Portfolio effects	Policy-mix integration	To what extent does the mission combine different support policy instruments to leverage their complementarities in achieving the mission goals? Has the mission been able to connect different policy instruments and regulations across the innovation chain?
	Consistency	Has the mission been able to assemble a set of complementary projects and activities across multiple dimensions to fulfil the mission? Has the mission promoted more interdisciplinary and/or intersectoral R&D?
	Exploratory	Has the mission been able to explore and select alternatives solutions to achieve the mission?
Material effects	Effectiveness	To what extent has the mission achieved the set objectives?
	Transformation	Is the mission contributing to a transformation of the sociotechnical system that underpins the challenge?